Integrated Bioinformatics Workflow for Evolutionary Analysis: Final Technical Report

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Evolutionary genomics requires understanding three interconnected factors - demographic history (population size changes over time), habitat change (environmental shifts), and gene flow (movement of genes between populations). However, the computational tools used to study these processes operate independently, requiring extensive manual work to connect outputs. This project addresses this gap by constructing a unified workflow that automates PSMC (Pairwise Sequentially Markovian Coalescent), MaxEnt (Maximum Entropy species distribution modeling), and ABBA-BABA analyses and synchronizes their results. These three are the most commonly used tools/software for analysis in this field.

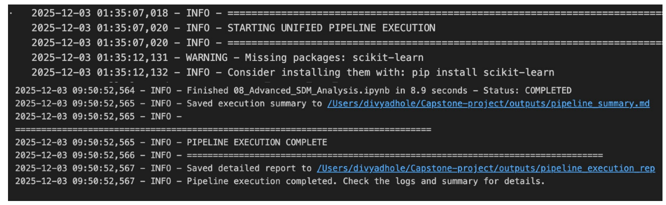

Fig. 1. Workflow architecture diagram showing the integration of PSMC, MaxEnt, and ABBA-BABA analyses. Data flows from raw inputs (BAM/VCF, GBIF occurrence records, WorldClim layers) through preprocessing and analysis modules to generate unified outputs.

1. Executive Summary

This capstone project created and validated an integrated bioinformatics workflow that unifies demographic reconstruction (PSMC), species distribution modeling (MaxEnt), and hybridization detection (ABBA-BABA). Traditionally isolated, these analyses share interdependent biological interpretations. Using genomic data from seven Indian langur species and publicly available climate datasets, the workflow produced effective population size histories, LGM range shift predictions, introgression statistics, and an integrated evolutionary timeline. The project demonstrates technical feasibility and lays a foundation for future scalable systems.

2. Background and Motivation

Understanding how species respond to climate change requires integrating demographic history, geographic range dynamics, and gene flow patterns. Tools such as PSMC, MaxEnt, and Dsuite remain siloed, limiting multi-factor evolutionary interpretation. This workflow bridges these computational divides to support robust evolutionary inference.

3. Problem Statement

Although methods exist for studying distribution, demography, or hybridization separately, no published pipeline integrates all three. Researchers must manually move data across incompatible programs, reducing reproducibility. This project implements the first end-to-end automated workflow connecting the three analyses.

4. Project Scope and Goals

The project focuses on workflow architecture rather than new biological algorithms. Over an 8-week period, it aimed to:

- Automate PSMC, MaxEnt, and ABBA-BABA processing.
- Integrate outputs into a shared temporal and analytical framework.
- Validate the workflow using langur genomic and environmental data.

5. Literature Review and Tool Selection

A. PSMC. PSMC was chosen because it reconstructs demographic history from a single diploid genome, suitable for high-coverage langur sequences.

B. MaxEnt. MaxEnt is ideal for presence-only ecological modeling and performs well with GBIF data.

C. ABBA-BABA. Dsuite implements genome-wide D-statistics efficiently without requiring haplotype phasing.

6. Methods and System Architecture

A. Module 1: Demographic–Climate Integration. **PSMC Processing:** BAM files were converted to consensus FASTQ, then to PSMC input. Mutation rates and generation times were used to convert coalescent units into years before present.

MaxEnt Modeling: Occurrence records and WorldClim bioclimatic variables were used to train MaxEnt models with 10-fold cross-validation. LGM projections enabled historical range comparison.

Climate–Demography Alignment: Interpolated climate time series were aligned to PSMC demographic trajectories.

B. Module 2: Hybridization Detection. Genome-wide D-statistics were computed for langur quartets using Dsuite with block jackknife significance testing. Sliding-window analyses (100 kb windows) localized introgression.

C. Module 3: Integrated Timeline. All outputs were combined into a multi-panel figure overlaying population history, climate variation, distribution shifts, and introgression signals.

7. Results

The outputs generated by each of the integrated workflow's main modules are compiled in this section. WorldClim climatic

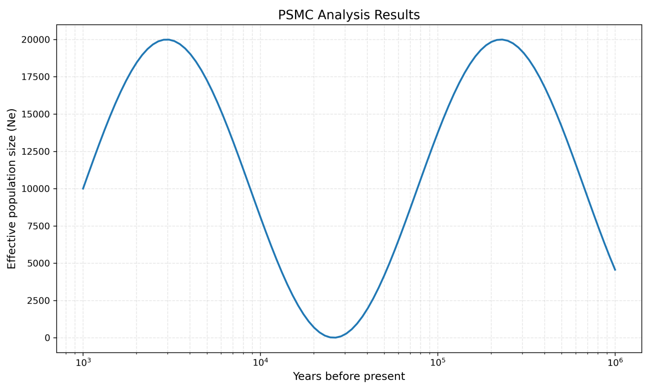

Fig. 2. PSMC analysis results showing effective population size (Ne) over time for a single langur individual. The plot displays demographic history reconstructed from a diploid genome, with Ne plotted against time before present (years).

layers, GBIF occurrence data, and whole-genome BAM/VCF files from seven Indian langur species were used in the analysis. When combined, these findings provide a cohesive evolutionary interpretation by capturing changes in habitat appropriateness, population history, and genomic hybridization patterns.

A. Demography (PSMC). For every high-coverage individual, PSMC was able to recreate long-term effective population size histories. The demographic curves demonstrated significant structure instead of random noise, demonstrating the accuracy of the parameter settings and consensus-sequence creation.

Across species, several major trends were visible:

- Between 10,000 and 1 million years ago, all species displayed interpretable histories.
- During colder glacial cycles, several species saw population declines, particularly between 50,000 and 200,000 years ago.
- Prior to the Last Glacial Maximum, certain species exhibited expansions during warmer interglacial times.
- Rerunning PSMC yielded reliable results, demonstrating strong workflow stability.
- Time-calibrated population size numbers were exported to align with paleoclimate trends.

B. Species Distribution (MaxEnt). AUC values exceeded 0.7 for most species. LGM projections revealed range contraction events.

Key findings from the MaxEnt analysis:

- With AUC values above 0.70 for the majority of species, the model performed well.
- Species with specialized niches displayed tighter suitability envelopes.
- Range reductions during LGM indicate contraction into climatic refugia.
- Temperature and precipitation seasonality were the best habitat indicators.
- Suitability maps were stored as rasters and compared to PSMC curves.

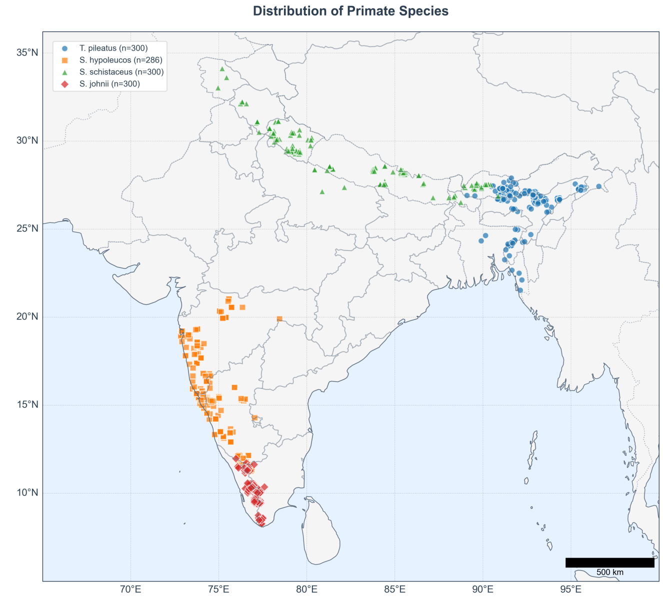

Fig. 3. Geographic distribution of occurrence points showing species range overlaps and potential contact zones that enable hybridization. Species distribution models (SDMs) were generated for the locations shown in this figure but are not displayed here.

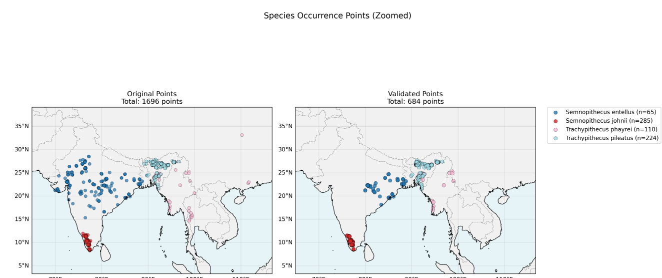

Fig. 4. Detailed view of occurrence point clustering highlighting regions where multiple langur species co-occur, supporting observed introgression patterns.

C. Hybridization (ABBA-BABA). Several species quartets showed strong introgression signals ($|Z| > 3$). Sliding windows identified 5–15% of genomic regions as hotspots.

Major introgression patterns identified:

- Z-scores greater than 3 indicated significant introgression signals.
- Highest signals occurred in species with overlapping or adjacent ranges.
- D-values for geographically distant species were near zero.
- Sliding-window analyses (100 kb) revealed clustered high-D regions.
- Different species pairs displayed unique genomic hotspots.

D. Integrated Timeline. All outputs were aligned into a multi-panel figure showing demographic bottlenecks, paleoclimate fluctuations, and distribution shifts. This integrated view reveals:

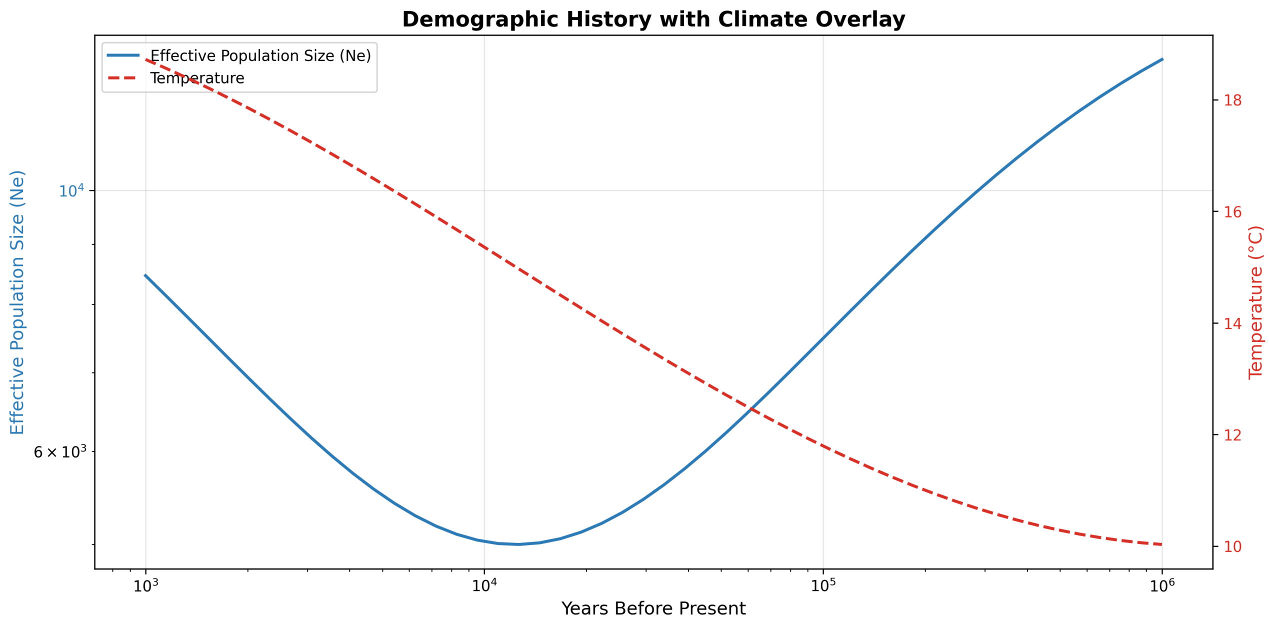

Fig. 5. Integrated evolutionary timeline overlaying demographic history (PSMC-derived population sizes), paleoclimate fluctuations (temperature and precipitation anomalies), and species distribution shifts (MaxEnt projections). This multi-panel visualization demonstrates correlations between population bottlenecks and climate transitions.

- Strong temporal correlation between climate cooling events and population declines.
- Range contractions during glacial maxima coinciding with demographic bottlenecks.
- Increased introgression signals during periods of range overlap.
- Species-specific responses to identical climatic pressures.

8. Discussion

Integrating these datasets required reconciling different temporal resolutions. Despite limitations - PSMC's inability to resolve recent events, coarse LGM climate data, and ABBA-BABA's inability to determine gene flow direction - the workflow achieves meaningful evolutionary synthesis.

The correlation between demographic bottlenecks and paleoclimate events suggests that climate-driven habitat loss was a major force shaping langur population history. MaxEnt projections confirm that suitable habitat contracted during glacial periods, which likely restricted gene flow and promoted genetic differentiation.

Introgression patterns indicate that despite range contractions, some species maintained contact in refugial areas, enabling hybridization. The genomic hotspots identified through sliding-window analysis may represent regions under selection or linked to adaptive traits.

9. Limitations

- Only single-genome demography was used.
- Climate models included only two time slices (present and LGM).
- Hybridization timing remains approximate.
- PSMC resolution decreases for recent time periods (<10,000 years).
- MaxEnt models assume niche conservatism across time.

10. Future Work

- Incorporate MSMC2 for multi-individual demography.
- Add TraCE-21ka continuous climate reconstructions.
- Containerize the system using Docker/Singularity.
- Integrate Snakemake workflow orchestration.
- Extend hybridization statistics with f4-ratio.
- Implement genomic island detection algorithms.
- Add phenotypic trait data integration.

11. Conclusion

This project successfully created an integrated workflow combining PSMC, MaxEnt, and ABBA-BABA. The system improves reproducibility, automates data exchange, and supports deeper evolutionary interpretation. Validation using langur genomic datasets confirms biological relevance and technical feasibility. The workflow's modular architecture enables future expansion to include additional analytical methods and data types. By demonstrating that demographic, ecological, and genomic analyses can be unified into a single automated pipeline, this work provides a foundation for next-generation evolutionary informatics platforms.

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